1. Fundamentals of Operations Management

- **Operations Management (OM):** Refers to the systematic design, direction, and control of processes that transform inputs into services and products for both internal and external customers.
- **Source of Advantage:** For top MNCs like Unilever, OM is a primary source of competitive advantage in both service and manufacturing sectors.
- **Key Definitions:**
    - **Process:** Any activity or group of activities that takes inputs, transforms them, and provides outputs for customers.
    - **Operation:** A group of resources performing all or part of one or more processes.
    - **Supply Chain:** An interrelated series of processes within and across firms that produce a service or product to satisfy customers.
    - **Supply Chain Management:** The synchronization of a firm's processes with those of its suppliers and customers to match materials, services, and information flow with customer demand.

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2. Role of Operations in an Organization

Operations management underlies all departments and functions within a business. It acts as a coordinator between the three mainline functions:

1. **Operations:** Transforms material and service inputs into product and service outputs.
2. **Marketing:** Responsible for producing sales revenue from outputs.
3. **Finance:** Generates financial resources and capital from investors and sales.

- **Integration:** Success depends on how well these unique functions work together toward a common strategy.
- **Support Functions:** Functions like Accounting, Information Systems, Human Resources, and Engineering provide essential managerial support.

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3. The Process View

A process view provides a realistic picture of how firms actually work.

- **Cross-Functional Nature:** While some processes are specific to one department (e.g., billing in accounting), many processes involve workflows that cut across departmental boundaries and require resources from several areas.
- **How Processes Work:** Every process has inputs and outputs.
    - **Inputs:** Include human resources (workers/managers), capital (equipment/facilities), materials, land, and energy.
    - **Outputs:** Can be services (often in the form of information) or tangible products.

- **Nested Processes:** A process can be broken down into smaller subprocesses, which may require different skills or be designed for routine vs. customized work.

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4. Manufacturing vs. Service Processes

While most firms perform both, there are key differences in how these processes are managed.

Key Differences

Characteristic	Manufacturing Processes	Service Processes
Output Nature	Physical and durable	Intangible and perishable
Inventory	Output can be inventoried	Output cannot be inventoried
Customer Contact	Low degree of contact	High degree of contact
Response Time	Long response times	Short response times
Intensity	Capital intensive	Labor intensive
Quality Measure	Quality is easily measured	Quality is not easily measured

- **Manufacturing Transformations:** These processes change materials in terms of physical properties, shape, size, surface finish, or joining parts.

- **Similarities:** Service providers often inventory their inputs (e.g., hospitals keeping medical supplies), and manufacturers often provide services (e.g., warranties and financing).

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5. The Supply Chain View

Strategic success requires processes to add value for customers throughout the entire supply chain. Processes are categorized into two main types:

A. Core Processes

Sets of activities that deliver value to external customers:

1. **Supplier Relationship Process:** Selecting suppliers and facilitating the flow of items into the firm.
2. **New Service/Product Development:** Designing and developing new products or services based on customer or market inputs.
3. **Order Fulfillment:** Activities required to produce and deliver the product or service.
4. **Customer Relationship Process:** Identifying, attracting, and building relationships with external customers (includes marketing and sales).

B. Support Processes

Essential resources and inputs provided to core processes. Examples include:

- **Human Resources:** Recruiting, training, and compensation.
- **Legal:** Ensuring compliance with rules and regulations.
- **Information Systems:** Movement and processing of data for business decisions.

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6. Operations Strategy

Operations strategy links long-term and short-term decisions to **Corporate Strategy**.

- **Corporate Strategy:** Provides the framework for all functions. It includes environmental scanning, identifying core competencies, and developing global strategies.
- **Core Competencies:** Unique strengths of an organization, including its workforce, facilities, market/financial know-how, and systems/technology.
- **Global Strategies:** Entering foreign markets through strategic alliances (collaborative efforts, joint ventures, or technology licensing) or by locating operations abroad.

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7. Competitive Priorities and Capabilities

Managers select specific dimensions to outperform competitors.

- **Competitive Priorities:** Operational dimensions a process *must* possess to satisfy customers.
- **Competitive Capabilities:** Dimensions a process *actually* possesses and delivers.

The Nine Competitive Priorities

Group	Priority	Definition
Cost	1. Low-cost operations	Delivering at the lowest possible cost.
Quality	2. Top quality	Delivering an outstanding service or product.
	3. Consistent quality	Meeting design specifications on a consistent basis.
Time	4. Delivery speed	Quickly filling a customer's order.
	5. On-time delivery	Meeting delivery-time promises.
	6. Development speed	Quickly introducing a new service or product.
Flexibility	7. Customization	Satisfying unique needs by changing designs.

	8. Variety	Handling a wide assortment of products efficiently.
	9. Volume flexibility	Accelerating or decelerating production rate quickly.

- **Time-Based Competition:** A strategy focusing specifically on delivery and development speed.

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8. Order Winners and Qualifiers

- **Order Qualifier:** The minimal level of performance required for a firm to even do business in a market segment. Fulfilling this only positions the firm to compete.
- **Order Winner:** A criterion that customers use to differentiate one firm's products from another's and make the final purchase decision. This can include price, quality, reputation, or technical support.

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9. Productivity

Productivity measures how effectively resources are used.

Factors Affecting Productivity

There are eight primary factors: Technical (technology/automation), Production (planning/quality), Organizational (authority/responsibility), Personnel (training/motivation), Finance (capital control), Management (efficiency), Government (rules), and Location (infrastructure/workforce).

Productivity Measures

1. **Labor Productivity:**
   Labor Productivity = Output / Labor Input
2. **Multifactor Productivity (MFP):** An index of output provided by more than one resource.
3. *MFP = Value of Output / (Labor Cost + Materials Cost + Overhead Cost)*

- **Productivity Growth:**
  Growth = [(New Productivity - Old Productivity) / Old Productivity] × 100%

Note: Relying only on labor productivity can be misleading; management should look at MFP for a more complete picture of process improvement.